



Deep learning for time series forecasting: a survey of recent advances

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Abstract

Time series forecasting plays a critical role in numerous real-world applications, such as finance, healthcare, transportation, and scientific computing. In recent years, deep learning has become a powerful tool for modeling complex temporal patterns and improving forecasting accuracy. This survey provides an overview of recent deep learning approaches for time series forecasting, involving various architectures including RNNs, CNNs, GNNs, transformers, large language models, MLP-based models, and diffusion models. We first identify key challenges in the field, such as temporal dependency, efficiency, and cross-variable dependency, which drive the development of forecasting techniques. Then, the general advantages and limitations of each architecture are discussed to contextualize their adaptation in time series forecasting. Furthermore, we highlight promising design trends like multi-scale modeling, decomposition, and frequency-domain techniques, which are shaping the future of the field. This paper serves as a compact reference for researchers and practitioners seeking to understand the current landscape and future trajectory of deep learning in time series forecasting.

Keywords

time series forecasting; deep learning; neural networks; survey; long term forecasting

1 Introduction

Time series forecasting is one of the classic time series analysis tasks that predicts the future pattern of time sequences based on the past trend. It plays a significant role across a wide range of application domains [1], from financial markets [2] and climate modeling [3] to healthcare monitoring [4], transportation [5], and scientific computing such as high-performance computing (HPC) system management [6]. In these fields, accurate prediction helps with informed decision making, efficient resource allocation, and proactive risk mitigation.

However, real-world time series forecasting task presents many significant challenges derived from the characteristics of the data. The data collected from real-world systems is often noisy, high-dimensional, non-stationary, and multivariate in nature [7–9]. Traditional statistical methods, such as autoregressive integrated moving average (ARIMA) and exponential smoothing (ETS), while effective in specific scenarios, typically fall short when dealing with long-term dependencies, nonlinear dynamics, or the scale and complexity of modern datasets [10].

In recent years, deep learning has emerged as a powerful tool in time series forecasting. Different types of models, such as recurrent neural networks (RNNs) [11], convolutional neural networks (CNNs)

[12], graph neural networks (GNNs) [13], and transformers [14] have demonstrated strong performance in both academic benchmarks and real-world applications. More recently, large language models (LLMs) [15] have been explored for their one-shot and zero-shot forecasting capabilities, and diffusion models (DMs) [16] have shown promise due to their strong generalization across diverse forecasting tasks. This evolution reflects a broader shift from task-specific architectures toward versatile models that can learn generalizable patterns, support low-shot forecasting scenarios, and adapt across diverse time series domains with improved interpretability [17–19].

This paper provides an overview of recent deep learning methods for time series forecasting. We organize our discussion around architectural categories, like RNNs, CNNs, GNNs, transformers, and other emerging paradigms, as well as key modeling trends such as frequency-domain methods, decomposition techniques, and efficiency.

The goal of this survey is to provide a concise yet up-to-date reference for researchers and practitioners interested in time series forecasting, particularly in the context of recent advancements in deep learning. Section 2 introduces the problem definition and fundamental concepts of time series forecasting. Section 3 then

examines key challenges that continue to drive innovation in this domain. Section 4 categorizes and reviews representative models developed in recent years, organizing them by architectural families. Section 5 distills common techniques and design patterns emerging from these models, highlighting shared strategies, novel ideas, and the specific challenges they help address. Finally, Section 6 discusses open questions and outlines promising directions for future research, aiming to inspire continued exploration at the intersection of model design and time series characteristics.

■ 2 Problem definition

2.1 Time series forecasting

A time series is an ordered sequence of observations recorded over time, where the temporal order is essential to the problem. Common examples include daily stock prices, hourly weather readings, and monthly sales data. Formally, a time series can be written as a sequence $(x_1, x_2, \dots, x_t)$, where x_i is the observed value at time step i .

Time series forecasting refers to the task of predicting future values based on past observations. Given historical data up to time t , the goal is to predict future values over a forecast horizon h , producing estimates for $(x_{t+1}, x_{t+2}, \dots, x_{t+h})$. Unlike conventional machine learning problems that often assume independent and identically distributed (i.i.d.) samples, forecasting models explicitly leverage the temporal dependencies among observations.

2.2 Forecasting settings

Time series forecasting tasks can be categorized based on the structure of the input data. In a univariate forecasting setting, the objective is to predict future values of a single variable based solely on its own historical observations. Formally, given a univariate time series

$$\mathbf{x} = (x_1, x_2, \dots, x_T)^\top \in \mathbb{R}^T, \quad (1)$$

and a lookback window of size T , the goal is to predict value(s) at a future time step or a sequence of future time steps.

In contrast, multivariate time series (MTS) forecasting involves multiple related time series evolving simultaneously. Let $\mathbf{x}_t \in \mathbb{R}^N$ denote the multivariate observation of N variables at timestamp t . The historical input at timestamp t is defined as the lookback window $\mathbf{X}_t = [\mathbf{x}_{t-T+1}, \mathbf{x}_{t-T+2}, \dots, \mathbf{x}_t] \in \mathbb{R}^{N \times T}$. The forecasting task aims to predict the next τ timestamps: $\mathbf{Y}_t = [\mathbf{x}_{t+1}, \mathbf{x}_{t+2}, \dots, \mathbf{x}_{t+\tau}] \in \mathbb{R}^{N \times \tau}$, based on the historical observations $\mathbf{X}_t$. This setting allows models to capture not only temporal dependencies within each individual series but also the cross-series interactions among multiple variables.

2.3 Forecasting horizons

Time series forecasting tasks can also be distinguished based on the forecasting horizon, that is, the number of future time steps to predict.

In single-step forecasting, the objective is to predict the immediate next value based on a fixed-size historical window. Formally, given the historical input $\mathbf{X}_t = [\mathbf{x}_{t-T+1}, \mathbf{x}_{t-T+2}, \dots, \mathbf{x}_t] \in \mathbb{R}^{N \times T}$, the goal is to learn a mapping function:

$$f: \mathbb{R}^{N \times T} \rightarrow \mathbb{R}^N, \quad (2)$$

such that the output $f(\mathbf{X}_t)$ estimates $\mathbf{x}_{t+1}$, the multivariate observation at the next timestamp, capturing temporal patterns in the input sequence.

In multi-step forecasting, the goal is to predict a sequence of future values over a forecast horizon of length τ . The corresponding mapping function is defined as:

$$f: \mathbb{R}^{N \times T} \rightarrow \mathbb{R}^{N \times \tau}, \quad (3)$$

where the output $f(\mathbf{X}_t)$ provides an estimate for the sequence $\mathbf{Y}_t = [\mathbf{x}_{t+1}, \mathbf{x}_{t+2}, \dots, \mathbf{x}_{t+\tau}] \in \mathbb{R}^{N \times \tau}$.

2.4 Forecasting outputs

Another important distinction in this field lies in the nature of the predicted outputs, which can be either single-point estimates or full probability distributions.

In point forecasting, the model aims to predict a single best estimate for each future time step. Formally, the function is defined as:

$$f: \mathbb{R}^{N \times T} \rightarrow \mathbb{R}^{N \times \tau}, \quad (4)$$

where $f(\mathbf{X}_t)$ produces a deterministic forecast for future sequence $\mathbf{Y}_t$.

In probabilistic forecasting, the model predicts a full probability distribution over future outcomes, capturing the inherent uncertainty in the forecasting task. The mapping is defined as:

$$f: \mathbb{R}^{N \times T} \rightarrow \mathcal{P}(\mathbb{R}^{N \times \tau}), \quad (5)$$

where $\mathcal{P}(\mathbb{R}^{N \times \tau})$ denotes the space of probability distributions over the forecast horizon. Probabilistic forecasts provide richer information by offering prediction intervals, quantiles, or full distributions, enabling more informed and risk-aware decision-making, especially in high-stakes applications where uncertainty plays a critical role [20].

■ 3 Challenges

Time series forecasting poses a diverse set of challenges that stem from the intrinsic characteristics of temporal data and the practical demands of real-world applications. As shown in Fig. 1, the research landscape has converged around several key challenges, which this survey categorizes into five major areas: temporal dependency, cross-variable dependency, data heterogeneity, efficiency, and uncertainty. These categories reflect the most pressing obstacles identified across recent literature. The figure illustrates the distribution of research attention among these categories, based on the papers reviewed in this survey. In the following subsections, we systematically outline each challenge category and its constituent issues, emphasizing their implications for deep learning-based forecasting task.

3.1 Temporal dependency

Capturing temporal dependency is fundamental to time series forecasting, as it governs how information evolves and relates over time. However, modeling these dependencies remains difficult due to the wide range of temporal dynamics present in real-world data,

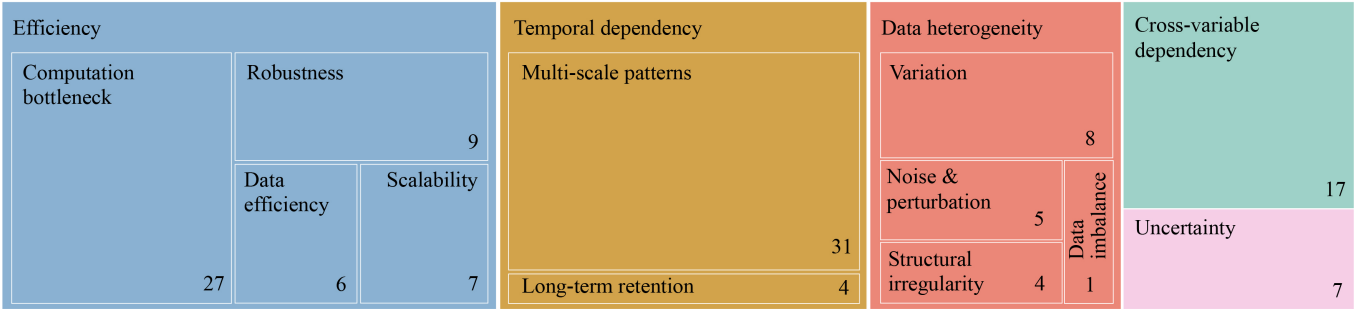


Fig. 1 Hierarchical categorization of challenges in time series forecasting. We visualize five high-level challenge categories: efficiency, temporal dependency, data heterogeneity, cross-variable dependency, and uncertainty, some of them are divided into representative sub-challenges. The relative size reflect the frequency and emphasis placed on each challenge across recent deep learning research we explore, providing a visual summary of key modeling concerns in the field

ranging from short-term fluctuations to long-term trends, and from periodic patterns to hierarchical structures. Deep learning models must effectively handle this diversity to achieve robust forecasting performance.

1) Multi-scale patterns

Temporal patterns often exist at multiple scales, such as rapid short-term changes, recurring periodic signals, and slowly evolving trends. These multi-scale patterns are difficult to capture within a single modeling resolution. Without explicit mechanisms to separate and represent different temporal granularities, models may fail to recognize important structure in the data. Additionally, hierarchical and nested temporal patterns further complicate learning, as they require the model to distinguish between overlapping and interacting temporal behaviors.

2) Long-term retention

A major challenge for recurrent models is retaining information over long temporal horizons. Classical RNNs face issues like gradient vanishing/exploding [21] and information forgetting issues [11], which weaken the influence of early time steps during backpropagation. Even advanced variants like LSTMs are limited by gating mechanisms that restrict memory updates, reducing their ability to adapt to evolving patterns. As a result, long-term dependencies are often lost, leading to poor performance on tasks that require extended temporal context.

In summary, temporal dependency poses fundamental obstacles to forecasting. It challenges models to retain meaningful information over time while also recognizing patterns across diverse and often nested temporal scales. These challenges become even more pronounced in the presence of noise, irregularities, or non-stationary dynamics. Without adequately addressing temporal dependency, predictive performance can degrade significantly, especially in long-horizon forecasting or tasks involving complex temporal structures.

3.2 Cross-variable dependency

In MTS forecasting, capturing the relationships between variables, also referred to as channels, is essential for fully leveraging the available information. However, modeling these cross-variable dependencies remains a significant challenge. Variables often interact

in complex, nonlinear ways that are difficult to capture through simple aggregation or independent modeling. These interactions may reflect latent structures or shared influences that, if not carefully modeled, can introduce noise or obscure meaningful correlations. A central modeling decision lies in whether to treat variables jointly or independently. Channel-dependent (CD) approaches aim to exploit inter-variable interactions to enhance predictive performance, offering greater representational power. However, they are more susceptible to overfitting, particularly when cross-channel signals are weak, noisy, or spurious. In contrast, channel-independent (CI) methods model each variable separately, which often leads to improved robustness but at the cost of potentially ignoring valuable inter-variable information [22].

Balancing these two paradigms: leveraging cross-variable information while avoiding overfitting remains a fundamental challenge in multivariate forecasting. Without mechanisms that can selectively and reliably capture inter-variable dependencies, model performance may degrade, especially in high-dimensional or noisy settings.

3.3 Data heterogeneity

Time series data in real-world applications is rarely uniform, its characteristics often vary significantly across sequences, variables, and domains. This heterogeneity manifests in multiple ways, including architectural mismatches, inconsistent variable semantics, sensitivity to noise, irregular sampling patterns, and imbalanced distributions. Each of these factors poses unique challenges for deep learning models. Understanding the different forms of heterogeneity is thus a foundational step toward building robust and scalable forecasting systems.

1) Variation

Variation is a pervasive challenge in time series forecasting, manifesting both at the architectural level, when applying models developed for other domains, and within the input data itself.

Many deep learning models, such as transformers and LLMs, were originally designed for natural language processing and struggle to align with the structural demands of time series data. For instance, transformers rely on permutation-invariant attention mechanisms and lack built-in temporal order modeling, which can lead to ineffective

token mixing and degraded performance in sequential tasks [23]. Similarly, LLM-based approaches face difficulties integrating continuous time series with discrete prompts, often yielding entangled or noisy representations [15]. Recent trends show a growing interest in adapting newly released models to time series forecasting, driven by their strong representational capabilities, even if they are not originally designed for this field. However, aligning such models with the unique structure and noise characteristics of temporal data remains a significant challenge.

At the data level, multivariate time series introduce substantial variation across variables. Each variable may exhibit distinct temporal dynamics, levels of noise, degrees of stability, and even differing semantic roles. However, many models apply uniform processing across all channels, assuming homogeneity that rarely exists in practice. This simplification can overlook critical inter-variable differences and obscure meaningful relationships, especially when variables are embedded into the same representation space [24]. When these variations occur in the series, effectively handling them remains a big challenge to solve.

Together, these challenges highlight the need for forecasting models that are both structurally adaptive and semantically aware, capable of aligning general-purpose architectures with the temporal domain while also respecting the nuanced differences between input dimensions.

2) Noise and perturbation

Real-world time series are frequently affected by noise from external disturbances, sensor inaccuracies, or irregular system behavior. Even minor fluctuations in the input can significantly disrupt learning, especially in attention-based models that may mistakenly assign importance to outliers [13]. This noise not only weakens temporal dependency modeling but also distorts inter-variable correlations, reducing the overall robustness of forecasting systems. Effectively managing noise and perturbations remains an open challenge, particularly for models applied across diverse and unpredictable environments.

3) Structural irregularity

Many real-world time series exhibit structural irregularities, such as uneven sampling intervals, missing observations, and non-uniform sequence lengths. These issues are particularly prevalent in domains like healthcare, finance, and environmental monitoring, where data is often collected asynchronously, intermittently, or under inconsistent conditions [25–27]. Such disruptions violate the core assumptions of traditional forecasting models, which typically require regularly spaced timestamps and complete input sequences. If not properly addressed, these irregularities can significantly impair forecasting accuracy and model robustness in practical applications.

Moreover, the presence of distributional shifts and evolving intra-series patterns adds further complexity to modeling. Irregularity in the temporal structure not only complicates data representation but also challenges the ability of models to capture consistent temporal dependencies. Addressing these irregularities requires models that are robust to missing data, adaptive to evolving dynamics, and capable of learning from temporally inconsistent inputs.

4) Data imbalance

A prominent challenge in real-world forecasting arises from the highly imbalanced nature of time series magnitudes, which often follow a heavy-tailed or power-law distribution [28]. For instance, in large-scale datasets like retail sales, the average weekly sales (or sales velocity) across millions of items can span several orders of magnitude. This scale-free property complicates efforts to group time series by value range or train separate models for specific magnitude bands, as each subgroup still exhibits similar skewness. Such imbalance also undermines the effectiveness of widely used normalization techniques, including input standardization and batch normalization, which rely on assumptions of more uniform distributions [29]. Moreover, group-based regularization approaches may fail to stabilize learning due to the vast variability within each group [30]. As a result, developing global forecasting models that generalize well across this magnitude imbalance remains an open and critical challenge.

In summary, data heterogeneity challenges the assumption that a single unified model can effectively learn from all time series equally. It exposes the limitations of generic architectural designs and highlights the need for models that are structurally flexible, semantically aware, and statistically robust. Addressing these diverse sources of variation is essential for enabling deep learning models to generalize reliably across real-world forecasting tasks.

3.4 Efficiency

Efficiency is a fundamental requirement for deploying deep learning models in real-world time series forecasting applications. Unlike offline benchmark scenarios, practical settings demand models that can deliver accurate predictions under constraints on computation time, memory usage, data availability, and deployment scale. However, current models often face significant challenges due to architectural complexity, instability during training, and the need for large volumes of high-quality data. These factors make it tricky to find a balance between predictive performance and practical usability.

1) Computation bottleneck

One of the core challenges in time series forecasting lies in the computational inefficiency of many deep learning architectures. For example, recurrent models are fundamentally limited by their sequential processing, which leads to high inference latency and excessive iteration counts when modeling long sequences. Transformer-based models, although widely adopted for their ability to model long-range dependencies through self-attention mechanisms, also suffer from significant computational drawbacks. Their memory and time complexity can scale quadratically with sequence length, leading to high training costs and substantial memory consumption.

Ultimately, many of the computational bottlenecks in time series forecasting stem from the architectural mechanisms designed to increase modeling power. Achieving high representational capacity often comes at the expense of speed and efficiency, making it necessary to carefully balance performance with computational cost.

2) Robustness

Another key challenge in time series forecasting is the lack of robustness in many deep learning models. Architectures with long memory paths or deep recurrence are often highly sensitive to hyperparameter choices, leading to instability during training. Even transformer-based models, despite their widespread success, have been shown to exhibit unstable optimization behavior, particularly when trained on limited data or in the absence of large-scale pretraining [31,32].

This instability is further exacerbated by the growing complexity and parameter count of modern architectures. As models scale up to improve expressiveness, their robustness can degrade, making them more prone to overfitting and less reliable on unseen data. The tension between model capacity and training stability presents a persistent barrier to practical deployment, especially in domains where data is scarce or distribution shifts are common.

3) Data efficiency

Many time series forecasting models are inherently data-hungry, requiring large volumes of clean and domain-specific data to train effectively. However, unlike fields such as natural language processing or computer vision, where massive, standardized datasets are readily available, time series datasets are often relatively small [33]. Publicly available collections for general-purpose time series modeling remain scarce, with most totaling only a few gigabytes. This lack of large-scale, diverse benchmarks limits the development of foundation models and hinders progress toward more generalizable forecasting solutions.

This lack of data poses serious challenges for training deep learning models, particularly when attempting to build models that generalize across tasks or domains. Without sufficient data, models are more prone to overfitting, brittle behavior, and poor performance in downstream applications. As discussed in the robustness subsection, training instability can be amplified when data is limited. Here, the challenge is not just stabilizing the model, but also using the limited data more effectively. Developing forecasting models that perform well in low-resource or few-shot settings remains an open and critical obstacle in the broader goal of generalizable time series forecasting.

4) Scalability

Scaling time series forecasting models across sequence length, dimensionality, and task diversity presents a significant challenge. Classical recurrent models, such as RNNs, are fundamentally constrained by their sequential processing nature, which leads to high iteration counts and unstable gradients when applied to long sequences. These limitations make them difficult to scale for long-term forecasting and large-scale applications. While more recent models, such as those incorporating hierarchical structures or extended attention mechanisms, have improved the ability to model longer contexts, they still struggle with high memory consumption, increased training time, and limited generalization across diverse forecasting tasks and domains. How to develop scalable architectures that balance efficiency, capacity, and generalizability remains an

open problem in the field.

The scalability challenge is not only about processing longer or larger inputs efficiently, but also about building forecasting models that maintain stable performance as context length, data dimensionality, and task complexity increase. As time series applications grow in scale and scope, addressing these limitations becomes increasingly essential.

In summary, improving efficiency in time series forecasting is a multifaceted challenge. It requires addressing the computational burdens of deep architectures, enhancing their robustness to instability, enabling generalization in low-data regimes, and ensuring scalability to long sequences and high-dimensional tasks. These issues are often deeply interconnected, and failure to manage any one of them can undermine the model's overall effectiveness. Solving the efficiency problem is essential not only for advancing the research frontier but also for making forecasting models viable in real-world environments.

3.5 Uncertainty

In many real-world forecasting scenarios, uncertainty is not just a byproduct, it is a central consideration. Forecasting systems are frequently used to support downstream decisions in domains such as finance, healthcare, and operations, where understanding the range of possible outcomes is as important as predicting the most likely one. However, most deep learning models are optimized for point forecasting and provide little to no insight into the confidence or variability of their predictions. This lack of uncertainty quantification poses a serious challenge, particularly in high-stakes or risk-sensitive applications. Real-world time series are inherently noisy and often influenced by unobserved or stochastic factors, making it difficult to rely on deterministic outputs alone. As a result, there is a growing need for forecasting models that can provide meaningful measures of uncertainty. Probabilistic forecasting is one direction that seeks to address this, but accurately estimating uncertainty, especially in multivariate and long-horizon settings, remains a challenging task.

■ 4 Deep learning architectures for time series forecasting

In this section, we categorize the representative models in recent years based on their backbone architectures, including RNNs, CNNs, GNNs, transformers, LLMs, DMs, and other emerging designs. For each category, we highlight the motivations behind its development, the architectural innovations introduced, and the representative models that reflect current trends in the field.

While earlier efforts often focused on adapting a single architecture to the forecasting task, recent work increasingly emphasizes hybrid designs and modular components tailored to specific challenges. To provide a concise overview of the relative strengths and limitations of each model family, Table 1 summarizes their main advantages and drawbacks, as reported in recent research. The following subsections provide a more detailed analysis of individual models within each category, illustrating how these architectures have evolved to address the diverse challenges of time series forecasting.

4.1 RNN-based models

RNNs are one of the foundational deep learning architectures for

Table 1 Pros and cons of model architectures for time series forecasting

Model	Advantages	Limitations
RNN	+ Sequential dependency modeling + Joint modeling of multiple series	– Limited long-term memory – Vanishing/exploding gradients – Forgetting past information
CNN	+ Local pattern extraction + Efficient parallel computation + Fast training and inference	– Limited long-range dependency modeling – Restricted receptive field – Requires stacking or dilation for global context
GNN	+ Variable-wise dependency modeling + Flexible graph structure	– High computational cost – Weak native temporal modeling
Transformer (Trans)	+ Captures long-range dependencies + Handles variable-length input flexibly + Modular and scalable design	– High memory and computation cost – May miss cross-variable correlations – Unstable training on small data
LLM	+ Unified modeling across diverse TS tasks + Strong generalization with minimal labels	– Input modality mismatch – Lacks temporal inductive bias
MLP	+ Low complexity + Fast training and inference	– Poor long-term dependency modeling – Information bottleneck
Diffusion Models (DM)	+ Strong generative capability + Flexible conditional/unconditional modeling + Effective uncertainty estimation	– High computational cost – Error accumulation (autoregressive models) – Slow inference (autoregressive models)

time series forecasting. They model sequential dependencies by updating a hidden state over time, enabling them to capture temporal patterns across series. However, traditional RNNs face challenges such as vanishing or exploding gradients and difficulty retaining long-term information over extended sequences.

To address these limitations, advanced RNN variants such as the Long Short-Term Memory (LSTM) [34] and Gated Recurrent Unit (GRU) [35] were introduced. These models use gating mechanisms to regulate the flow of information and enable more stable learning of long-range dependencies. Nonetheless, the need for sequential processing and sensitivity to long-range dependencies have contributed to the growing dominance of more scalable alternatives such as transformers [36,37]. Despite this, some recent studies revisit the RNN paradigm with innovations aimed at overcoming these historical challenges.

DeepAR [28] is an RNN-based model designed for probabilistic forecasting. Instead of training separate models for each series, DeepAR learns a global model using an LSTM architecture, enabling it to capture shared patterns and handle series with limited history. It addresses challenges like highly skewed magnitude distributions by modeling the output with flexible likelihood functions such as the negative Binomial. DeepAR produces calibrated probabilistic forecasts with the help of feature engineering, allowing accurate quantile estimation. Its ability to learn complex seasonal and covariate effects with minimal manual effort makes it a strong choice for uncertainty-aware forecasting tasks.

WITRAN (Water-wave Information Transmission and Recurrent Acceleration Network) [11] is inspired by the idea that RNN limitations can be mitigated by dividing input sequences into sub-series and processing them in a structured fashion. The model introduces a bi-granular information transmission framework that rearranges the sequence into a two-dimensional structure based on its natural period, allowing information to propagate across both intra- and inter-periodic dimensions. At its core is the Horizontal-Vertical Gated Selective Unit (HVGSU), which performs gated recurrence

along both axes to capture local and global temporal dependencies. WITRAN also includes a Recurrent Acceleration Network (RAN) to reduce computational overhead, achieving strong long-horizon forecasting accuracy with better scalability than attention-based counterparts.

SegRNN [38] also revisits the limitations of RNNs in long-term forecasting, attributing their performance degradation to the excessive number of recurrent iterations resulting from extended look-back and forecast horizons. It introduces segment-wise recurrence during encoding to reduce the number of recurrent steps and employs a Parallel Multi-step Forecasting (PMF) strategy in decoding. PMF replaces autoregressive decoding with simultaneous prediction of all future steps using positional embeddings, which helps avoid error accumulation. By reducing sequential depth and optimizing the decoding process, SegRNN improves both forecasting accuracy and runtime efficiency.

While most recent advances focus on improving gradient-based training, a less-known class of recurrent neural networks, Echo State Networks (ESNs) [39], take a non-gradient-based approach to forecasting by relying on a fixed, randomly initialized recurrent reservoir and training only a readout layer via ridge regression. This avoids the vanishing gradient problem and enables fast training, but traditional ESNs suffer from high computational complexity and unstable dynamics in long-memory scenarios. LCESN (Locally Connected Echo State Networks) [40] extends this framework by adopting a locally connected reservoir structure, reducing the number of interactions and enabling larger, more efficient reservoirs. It also introduces a forced memory mechanism that feeds each neuron a delayed version of its own activation, enhancing the model's ability to capture long-range dependencies without backpropagation. Together with online update capabilities, LCESN shows that non-gradient-based recurrent models remain viable for efficient time series forecasting.

P-sLSTM [41] revisits recurrent neural networks for time series forecasting by enhancing the sLSTM architecture [42] with patching

and channel independence. While standard LSTMs struggle with long-term dependencies and sLSTM improves memory revision through exponential gating, P-sLSTM addresses residual short-memory issues by segmenting input sequences into patches, allowing the model to better capture global patterns. Additionally, channel independence is introduced to prevent overfitting and reduce complexity, a design rarely explored in RNNs. P-sLSTM demonstrates strong empirical performance, outperforming LSTM and sLSTM in most settings and achieving competitive results compared to transformers and MLP-based models. Its lower time complexity and robustness to overfitting make it a practical alternative for long-horizon multivariate forecasting tasks.

In summary, while RNNs have inherent limitations in long-term memory and training stability, recent research has shown that these issues can be mitigated through architectural improvements. Models like WITRAN and SegRNN improve scalability and reduce the burden of deep recurrence, while LCESN revitalizes the efficiency of non-gradient-based approaches. These advances show that well-designed RNN-based models can still deliver strong performance.

4.2 CNN-based models

CNNs have become a popular architecture in time series forecasting due to their ability to efficiently extract local temporal patterns through convolutional filters [43]. Unlike RNNs, CNNs process inputs in parallel across time steps, enabling fast training and inference while avoiding gradient instability [44]. These properties make CNNs particularly suitable for long sequences and resource-limited environments. However, standard CNNs are inherently limited in capturing long-range dependencies due to their fixed receptive fields. Expanding the receptive field typically requires stacking many layers or using dilation, leading to extra complexity.

Temporal Convolutional Networks (TCNs) [45] are specifically designed to model temporal dependencies by extending traditional CNNs with dilated and causal convolutions. Dilated convolutions expand the receptive field efficiently, while causal convolutions preserve the order of time steps. TCNs are typically built with residual blocks of dilated causal layers and are widely used in time series tasks for their efficiency and strong temporal modeling ability.

While TCNs improve the modeling of longer sequences, their use of shared convolutional filters and limited receptive fields in the intermediate layers constrain their expressiveness. Motivated by the observation that temporal relationships such as trends and seasonality are largely preserved after downsampling, SCINet (Sample Convolution and Interaction Network) [46] addresses these issues with a hierarchical architecture that processes downsampled time series through separate branches. By recursively splitting inputs and applying distinct convolutional filters, SCINet captures diverse temporal features. The SCI-Block enables cross-branch interaction, and the overall binary-tree structure with residual connections allows the model to integrate short- and long-term dependencies efficiently.

MICN (Multi-scale Isometric Convolution Network) [47] enhances temporal forecasting by jointly modeling local dynamics and global structures. It decomposes inputs into seasonal and trend-cyclical components and applies multi-branch convolutional modules to

extract complementary features. These include local convolutions for fine-grained changes, global convolutions for broader context, and isometric convolutions for multi-scale fusion. A merge operation is finally adopted to fuse the information. The design achieves linear complexity in both time and space, reducing many unnecessary computations.

TimesNet [12] reformulates time series as 2D tensors to better capture complex temporal dependencies driven by multi-periodicity. In this Temporal 2D-Variation framework, intra-period dependencies refer to short-term patterns within a single period (e.g., daily cycles), while inter-period dependencies reflect long-term trends across different periods (e.g., weekly or yearly variations). By separating these along two spatial axes, the model explicitly disentangles layered temporal structures. The TimesBlock module applies inception-style convolutions to extract multi-scale features from this 2D representation. Periods are selected via Fourier analysis, and outputs are aggregated using normalized frequency responses. This design enables the model to have strong generalizability, achieving convincing results in many mainstream analysis tasks, like forecasting, imputation, classification, and anomaly detection.

TVNet [48] further explores temporal dependencies by structuring inputs as 3D tensors to jointly model intra-patch, inter-patch, and cross-variable patterns. Each convolutional block uses shared base kernels modulated by learned adaptive weights to adjust to different temporal contexts. This structure allows the model to remain efficient while capturing complex dependencies across time and variables, demonstrating strong performance across multiple tasks.

From the exploration, people can see that modern CNN-based models have evolved beyond standard 1D convolutions by incorporating hierarchical decomposition, multi-branch architectures, and reformulated representations such as 2D and 3D tensors [9]. These developments improve the modeling of long-range dependencies while preserving the efficiency and parallelism inherent to CNNs. Despite their limited receptive fields and need for architectural extensions to capture global context, CNN-based approaches continue to offer a compelling balance between accuracy and efficiency for time series forecasting.

4.3 GNN-based models

GNNs have gained attention in time series forecasting for their ability to model structured dependencies among variables. Unlike sequence-based models that assume fixed temporal ordering, GNNs treat each variable as a node and learn variable-wise correlations through message passing on a graph structure. This formulation enables flexible modeling of multivariate time series and remains effective even in challenging scenarios such as irregular sampling or missing values. GNNs excel at capturing inter-variable interactions through adaptive or learned topologies. However, they generally lack native mechanisms for temporal modeling and often require integration with separate modules [49,50]. Additionally, spectral or dynamic graph operations can introduce high computational overhead, especially when applied to long sequences or high-dimensional data.

Early GNN-based time series models typically used static or

predefined graphs to capture spatial relationships among variables [51–53]. While effective in some settings, they failed to capture evolving patterns or task-specific dependencies. Recent research focuses on dynamic graph learning and joint modeling of intra- and inter-series correlations, converting the problem into a structure-aware reasoning task.

FourierGNN [54] reimagines time series forecasting purely as a graph learning problem, without relying on explicit temporal encoders. Unlike prior models that combine GNNs with temporal modules, FourierGNN represents each multivariate time series as a fully connected graph with nodes for time steps and edges for variable-wise relationships. It applies spectral graph convolutions with learnable filters in the Fourier domain, allowing the model to accommodate adequate learning expressiveness while achieving much lower complexity.

CrossGNN [13] is motivated by the observation from [23] that simple linear models can outperform many state-of-the-art forecasting architectures. This counterintuitive result prompts an investigation into why models that incorporate cross-time and cross-variable interactions do not consistently offer superior performance. The authors attribute this to noise in real-world MTS, which can weaken temporal signals and corrupt variable relationships. CrossGNN introduces an Adaptive Multi-Scale Identifier (AMSI) to extract stable temporal features in the frequency domain, and constructs separate GNN modules for cross-time and cross-variable reasoning. These modules selectively propagate information using correlation- and saliency-based filters, improving robustness to noise and enhancing long-horizon forecasting accuracy.

T-PatchGNN [55] targets irregularly sampled multivariate time series (IMTS), which pose unique challenges such as asynchronous measurements and non-uniform sequence lengths. The model segments univariate inputs into variable-length patches with uniform resolution, preserving local temporal and semantic structure. A time-aware convolutional encoder extracts features from each patch, while intra-series dependencies are modeled using transformers. Inter-variable interactions are captured by dynamically constructing time-varying graphs over patches, enabling scalable and accurate forecasting in complex domains like healthcare and climate analysis.

GNN-based models offer a flexible and expressive framework for modeling variable-wise dependencies in time series data. By leveraging adaptive topologies, spectral reasoning, and patch-level abstractions, recent models expand the capability of GNNs beyond static graph reasoning. Although GNNs typically require auxiliary components to model temporal dynamics and can be computationally intensive, their strength in capturing structured inter-variable relationships makes them a powerful tool for multivariate time series forecasting.

4.4 Transformer-based models

Transformer architectures have become increasingly dominant in time series forecasting due to their ability to model long-range dependencies, handle variable-length inputs, and support modular, scalable designs. Originally introduced for natural language processing (NLP) [56], transformers were adapted to time series

through innovations such as sparse attention, decomposition-based inductive biases, and frequency-domain representations [37,57,57]. These early developments demonstrated the effectiveness of attention mechanisms in capturing complex temporal patterns, particularly for long-horizon forecasting.

Despite their success, transformers face several known challenges in time series applications. Their quadratic complexity in sequence length leads to high memory and computational costs. In multivariate settings, standard attention mechanisms may struggle to capture cross-variable interactions effectively. Additionally, training transformers on small or noisy datasets often results in unstable optimization and poor generalization. This subsection highlights recent transformer-based models that go beyond those covered in the detailed survey by [58], focusing on newly released architectures that further adapt transformers to the demands of time series forecasting.

PatchTST [59] introduces patch-based tokenization for time series forecasting, where consecutive time steps are aggregated into fixed-length patches with stride S , allowing for richer local semantics and reduced sequence length. This improves locality modeling and reduces the computational complexity compared to point-wise tokenization. PatchTST also adopts a channel-independent attention mechanism, applying self-attention separately to each channel to avoid interference from weakly correlated variables. This strategy, previously effective in CNNs and linear models, proves beneficial for transformers as well. It further supports self-supervised pretraining via masked patch reconstruction, enabling strong representation learning. It achieves great performance in both supervised and self-supervised settings, highlighting its efficiency and expressiveness for multivariate forecasting.

Crossformer [60] targets a key limitation in transformer-based multivariate time series forecasting: the lack of explicit cross-dimension dependency modeling. Standard transformers embed all dimensions at each time step into a single token, capturing temporal patterns but overlooking inter-variable relationships. Crossformer addresses this by introducing Dimension-Segment-Wise (DSW) embedding, segmenting each series and forms a 2D representation with separate time and dimension axes. It then applies a Two-Stage Attention (TSA) mechanism to independently model cross-time and cross-dimension dependencies. These components are integrated into a Hierarchical Encoder-Decoder (HED) that captures multiscale patterns. Experiments show that Crossformer achieves strong performance on long-horizon MTS forecasting, validating the benefit of explicit dimension-aware modeling.

CARD (Channel Aligned Robust Blend Transformer) [61] revisits the trade-off between channel-dependent and channel-independent modeling in multivariate forecasting. The model introduces Channel Aligned Attention that aligns temporal tokens across variables to enable structured cross-channel learning, paired with a Robust Blend Module that fuses local and global context through lightweight token integration. This combination enables the model to balance robustness and capacity, offering a structured approach to effectively model inter-channel dependency in multivariate forecasting.

iTransformer [24] rethinks the limitations of transformer-based

forecasting in light of recent findings [62] showing that simple linear models can outperform more complex attention-based architectures in time series tasks. The authors argue that transformers are not inherently ineffective for time series, but rather misused due to inductive biases misaligned with temporal forecasting. To address this, iTransformer proposes an inverted architecture that shifts the attention operation from the variable dimension to the temporal dimension, performing self-attention independently within each channel across time. This design enables channel-specific temporal modeling, mitigates interference from weakly correlated variables, and reduces parameter redundancy.

To capture multi-scale temporal dependencies more effectively, Pathformer [14] introduces a multi-path transformer framework with adaptive routing. Each path corresponds to a different temporal scale, such as short-term, mid-term, or long-term dependencies, and processes input tokens using a specialized attention configuration. A lightweight routing controller dynamically selects the most suitable path for each token based on its contextual properties, like trend and seasonality, allowing the model to perform adaptive scale-aware modeling. This architecture enables effective specialization across temporal resolutions while maintaining the flexibility to fuse information across scales.

SAMformer [63] focuses on the training instability of transformers, particularly in data-scarce environments where overparameterized attention layers tend to converge to sharp local minima [31–32]. It incorporates Sharpness-Aware Minimization (SAM) into training procedure, which encourages convergence to flatter loss regions. The model also integrates reversible instance normalization [64] and channel-wise attention [65–66], design choices adopted from computer vision [67,67] that enhance numerical stability and variable sensitivity. SAMformer achieves stable and expressive forecasting with a shallow transformer architecture.

Timer-XL [68] generalizes the training objective of language modeling to multivariate time series forecasting by casting the task as next-token prediction across all variables. This formulation enables a unified modeling approach that naturally accommodates univariate, multivariate, and covariate-informed inputs. It introduces TimeAttention, a causal self-attention mechanism tailored for time series data. The design disentangles temporal and variable dependencies through a Kronecker-structured attention mask and permutation-equivariant positional embeddings, ensuring both causality and variable-invariance. This setup allows the model to scale to long sequences and make information-complete predictions across time and variables. By pretraining on large-scale datasets, Timer-XL also outperforms recent large models in zero-shot forecasting.

Recent transformer-based models extend beyond scaling attention to better align with the characteristics of time series data. They introduce solutions to mitigate computational cost, stabilize training, and improve cross-variable reasoning. These developments show that with appropriate architectural and optimization strategies, transformers can serve as flexible and general-purpose forecasters for long-range, multivariate, and context-rich time series scenarios.

4.5 LLM-based models

LLMs have recently attracted growing interest in time series forecasting due to their ability to handle a wide range of tasks through prompt-driven generalization. Their success in NLP stems from large-scale pretraining on massive text corpora, which equips them with strong generalization capabilities, contextual understanding, and few-shot adaptability. These strengths have motivated researchers to explore whether LLMs can similarly generalize across diverse time series tasks. Unlike traditional models that are typically tailored to specific time series problems, many LLM-based approaches aim to unify tasks by converting temporal inputs into token sequences compatible with pretrained LLMs. This paradigm enables broad generalization with minimal labeled data and offers a scalable solution in low-resource or cross-domain settings.

However, LLMs are not naturally designed for continuous, temporally structured data. Their lack of temporal inductive bias and reliance on discrete token inputs introduce critical challenges when applying them to time series. Notably, standard tokenization schemes fragment continuous numerical values into inconsistent and often non-meaningful sub-tokens, leading to the loss of fine-grained temporal granularity. Repetitive patterns, typical in time series, are disrupted by arbitrary token splits, while timestamps and sensor values are broken into multiple tokens without preserving their temporal semantics [69]. This can obscure short-term dynamics, hinder the capture of micro-trends, and result in unreliable predictions, especially in high-frequency or volatile signals. Addressing these pitfalls requires careful design choices, such as improved tokenization strategies or hybrid modeling approaches, to enable LLMs to effectively process time series data.

OneFitsAll (OFA) [70] addresses the limited availability of large-scale time series data by reusing pretrained LLMs as universal sequence learners. Rather than retraining or modifying model architectures, OFA reformulates time series tasks using learnable prompts that wrap inputs into formats interpretable by frozen LLMs. This enables task distinctions to be encoded directly in prompt tokens and allows for efficient transfer to multiple time series tasks.

TEMPO [71] combines decomposition-based modeling with prompt-based generation. Historical sequences are converted into structured prompts that include target lengths and temporal context, and are fed into a decoder-only transformer. To guide learning, the model applies additive decomposition to split the input into trend, seasonal, and residual components, each encoded using component-aware prompts. This hybrid design helps the model capture domain-specific patterns while remaining flexible across different datasets and tasks.

Time-LLM [15] tackles the fundamental challenge of adapting large language models to time series forecasting, a domain characterized by continuous inputs and temporal structures misaligned with the discrete token-based reasoning LLMs are trained on. It introduces a reprogramming framework that transforms time series data into prompts for autoregressive decoding by frozen LLMs. A learnable reprogramming layer transforms historical sequences into a structured prompt format that allows the model to

treat forecasting as conditional sequence generation. Without any fine-tuning or architectural change, the model performs accurate conditional forecasting by learning to map continuous sequences into discrete language-compatible formats.

TimeCMA [72] builds on the observation that directly prompting LLMs with time series inputs often leads to entangled, suboptimal representations. The authors propose a cross-modality alignment framework with two encoding branches: one for raw time series, and another for textual prompts processed by a frozen LLM. A similarity-based retrieval module aligns the two embedding spaces and selectively extracts channel-wise information from the LLM's prompt outputs to enhance time series representations. This design allows the model to benefit from language priors while preserving structural interpretability.

LLM-based models show a shift toward foundation-style forecasting by treating time series problems as general sequence modeling tasks. They offer strong generalization and can unify multiple tasks within a single framework. However, they also face challenges due to the mismatch between language and time series data, the lack of built-in temporal structure, and tokenization-induced loss of fine-grained temporal information. A growing research direction explores time series reasoning as forecasting, where LLMs are not only used for point predictions but also to perform reasoning over temporal patterns, such as explaining trends, identifying anomalies, or uncovering latent causal relationships. This contrasts with traditional statistical or deep learning methods, which typically lack innate reasoning capabilities. Continued progress in disentangled representations, reprogramming strategies, modular adaptations, and reasoning-oriented prompt designs may unlock scalable, interpretable, and task-agnostic forecasting solutions driven by pretrained language models.

4.6 MLP-based models

While attention-based models have dominated recent research in time series forecasting, MLPs have reemerged as a competitive alternative, due to their architectural simplicity, fast training, and low computational cost [17,73]. Their feedforward structure allows for high parallelism and efficient deployment, particularly in large-scale or long-horizon scenarios. However, standard MLPs struggle to model long-term dependencies and often suffer from information bottlenecks, limiting their effectiveness in capturing global and hierarchical temporal patterns. Enhancing the temporal modeling capacity of MLPs while preserving their inherent efficiency has therefore become a key area of innovation.

TSMixer [74] challenges the assumption that multivariate deep models always outperform univariate ones in time series forecasting. Motivated by findings that simple univariate models often surpass transformer-based multivariate models on common benchmarks, TSMixer builds on stacked temporal linear layers and introduces a lightweight MLP architecture that alternates between time-mixing and feature-mixing operations. This structure retains the strengths of linear models while allowing controlled use of cross-variate information. TSMixer performs on par with leading univariate

models and surpasses other multivariate models on standard datasets. TSMixer also shows strong gains on the M5 benchmark, where cross-variable signals are essential, confirming its effectiveness.

FreTS [75] introduces frequency-domain modeling to address the limitations of point-wise mappings in MLPs. It applies the Discrete Fourier Transform (DFT) to convert time series inputs into the frequency domain, where two parallel MLP modules extract temporal and cross-variable dependencies. This approach takes advantage of the compactness and global view offered by spectral representations. The processed signals are then projected back to the time domain using an inverse DFT. By combining global structural awareness with the efficiency of feedforward networks, FreTS improves long-sequence forecasting while maintaining simplicity and interpretability.

SOFTS [76] introduces an efficient MLP-based architecture for MTS forecasting, providing a scalable alternative to attention-based models. Its core component, the STar Aggregate-Redistribute (STAR) module, replaces pairwise channel interactions with a centralized mechanism that aggregates global information into a shared core and redistributes it to each local series. This design enables indirect channel interaction with linear complexity, reducing computational cost and improving local correlation. Besides, SOFTS scales effectively to time series with a large number of channels or long input sequences, a setting where transformer-based models often struggle without specialized modifications. Empirically, SOFTS matches or outperforms prior models while requiring significantly fewer computational resources, and STAR can also serve as a general drop-in replacement for attention layers in other forecasting architectures.

Motivated by the observation that time series exhibit distinct temporal behaviors at different resolutions, TimeMixer [77] focuses on disentangling temporal patterns across resolutions. Its two-stage design separates past extraction and future prediction using multi-scale mixing modules tailored for seasonal and trend components. By fusing outputs from different temporal levels, TimeMixer effectively captures both short- and long-term dynamics without increasing model complexity.

AMD (Adaptive Multi-Scale Decomposition) [78] further modifies the decomposition techniques used by prior MLP-based model, which fail to represent complex and overlapping temporal dynamics effectively [79]. It introduces hierarchical decomposition and patch-wise dependency modeling through a modular pipeline of decomposable mixing, dual interaction blocks, and adaptive fusion. The result is a lightweight but expressive model capable of learning rich temporal structures across multiple variables.

Recent MLP-based models have advanced considerably through the use of principled architectural enhancements that address historical shortcomings in long-range modeling. By integrating different techniques to enhance their ability to capture global dependencies, MLPs could achieve strong performance while maintaining their original low complexity and fast training. These developments reaffirm the potential of MLPs as an efficient backbone for this field.

4.7 Diffusion-based models

Diffusion models, originally developed for high-quality image and audio generation, have recently been adapted to time series forecasting due to their ability to model complex data distributions and generate diverse, probabilistic outputs. By learning to reverse a gradual noise-injection process, diffusion models enable flexible conditional and unconditional generation, making them well-suited for uncertainty-aware forecasting. Unlike deterministic models that produce point estimates, most diffusion-based approaches model the full distribution over future trajectories. This allows for richer uncertainty estimation and the generation of multiple plausible futures. However, existing diffusion models often incur high computational costs, particularly in autoregressive designs, which suffer from slow sampling and error accumulation when modeling long sequences.

Early efforts such as TimeGrad [80] introduced autoregressive denoising diffusion models to learn the conditional distribution of each future time step given past observations and known covariates. By combining a RNN encoder with a conditional diffusion process, TimeGrad models each future time step probabilistically. D3VAE [81], on the other hand, proposed a generative framework that integrates a coupled diffusion model with a bidirectional variational autoencoder (BVAE), enabling joint modeling of both input and output time series. It further incorporates a denoising score-matching network and latent disentanglement to clean diffused target time series and improve interpretability of temporal patterns. These early models demonstrated the potential of diffusion-based approaches in enhancing both the accuracy and reliability of probabilistic forecasting.

More recently, TSDiff [16] introduces an unconditional diffusion framework that supports forecasting, refinement, and data synthesis without retraining. Instead of conditioning the model during training, TSDiff introduces a self-guidance mechanism at inference time to help generation based on observed values. This design allows the model to support diverse scenarios, including probabilistic forecasting via guided sampling, plug-and-play refinement of external predictions using the learned prior, and generation of realistic synthetic data to support downstream training. Built on stacked S4 layers [82], TSDiff enables flexible downstream adaptation with realistic sample generation.

Motivated by the fast generation of Latent diffusion models [83], LDT (Latent Diffusion Transformer) [84] extends this paradigm to the domain of multivariate time series forecasting, addressing the challenge of high-dimensional sequence generation. LDT adopts a two-stage design that separates representation learning and generative modeling. In the first stage, it compresses time series into a latent space using dynamic normalization to handle distribution shifts. In the second stage, a non-autoregressive diffusion transformer performs denoising in the latent space, guided by self-conditioning and classifier-free mechanisms to balance flexibility and control. This design improves scalability and expressiveness in high-dimensional settings.

ARMD (Auto-Regressive Moving Diffusion) [85] rethinks

diffusion-based forecasting by aligning the generative mechanism with the intrinsic sequential dynamics of time series data. It aligns the diffusion process with sequential dynamics by introducing a chain-based formulation inspired by classical ARMA models. Instead of starting from noise, it generates forecasts through a sliding deterministic process and a distance-aware devolution network. This formulation treats the sampling process itself as the prediction trajectory, unifying the modeling of stochasticity and autoregressive progression within a single framework, enhancing interpretability and predictive stability.

D3U (Diffusion-based Decoupled Deterministic and Uncertain framework) [86] proposes separating deterministic components from residual uncertainty. From empirical case studies, the authors find that existing point forecasting models are often more effective at modeling trend and seasonal components than residuals, which tend to be highly uncertain. They apply a pre-trained point forecasting model captures deterministic structures and provides strong predictions along with informative representations of high-certainty components. These representations are then used as conditional context for a diffusion-based denoising model, which focuses exclusively on modeling residual uncertainty. This two-part design simplifies uncertainty modeling and improves interpretability, especially for multivariate forecasting.

Diffusion-based models provide a powerful probabilistic alternative to traditional forecasting approaches. Recent studies have enhanced their practicality by reducing computational overhead, mitigating autoregressive inefficiencies, and improving alignment with the structure of time series data. As more mature strategies emerge for applying these generative models, diffusion-based approaches are increasingly seen as a promising foundation for general-purpose, uncertainty-aware forecasting in high-dimensional and complex domains.

4.8 Other architectures

Beyond the dominant deep learning backbones, several recent models have explored alternative architectural principles inspired by some new architectures or different algorithms. These approaches often combine classical modeling insights with targeted deep learning components to improve the performance in time series forecasting.

FiLM [87], inspired by recursive memory models [88], enhances long-range forecasting by preserving historical information while filtering out noise. It begins by encoding long-term memory using orthogonal polynomial bases without learnable parameters. To remove high-frequency noise from the compressed representation, FiLM applies a Frequency Enhanced Layer (FEL) that filters high-frequency noise using low-frequency Fourier components and low-rank approximation. A lightweight mixture-of-experts component is further used to capture multi-scale temporal patterns. Designed to be modular and efficient, it performs well on both univariate and multivariate forecasting tasks.

TS-LIF [89] explores the potential of Spiking Neural Networks (SNNs) for time series forecasting, motivated by their biologically inspired temporal encoding and low-energy computation [90]. It leverages spiking neurons' inherent temporal encoding and energy

efficiency, while addressing traditional SNN limitations, such as weak memory and limited multi-scale representation, through a dual-compartment design. This structure separates dendritic and somatic processing, allowing the network to encode both low- and high-frequency dynamics. The resulting model exhibits robustness in sparse or low-resource scenarios.

TimeMachine [91] introduces a context-aware forecasting architecture that captures long-range dependencies in MTS using purely State Space Models (SSMs). Built upon the Mamba SSM [92] as its core inference engine, TimeMachine maintains linear scalability and low memory usage while modeling both global and local contexts. The model operates across two temporal resolutions via downsampling, extracting high- and low-resolution cues using distinct SSM modules. It unifies channel-mixing and channel-independent forecasting by deploying four SSMs to handle both global interactions and localized signals across channels. This design allows TimeMachine to selectively attend to relevant temporal content at multiple scales. Empirical results demonstrate that it achieves strong accuracy and efficiency across standard benchmarks, establishing SSMs as a competitive alternative to attention-based architectures.

KooNPro [93] builds on Koopman operator theory [94], which represents nonlinear dynamics through linear decompositions using eigenfunctions. While powerful, traditional Koopman methods can be sensitive to perturbations. KooNPro addresses this by introducing a pseudo-spectral framework that captures continuous latent variations and improves robustness. It also incorporates Neural Processes to model global structure and enable probabilistic interpolation. By integrating Koopman-based latent dynamics with distributional modeling, KooNPro produces interpretable and uncertainty-aware forecasts, even under sparse or noisy conditions.

DLinear [23] and LTBoost [95] revisit classical linear forecasting as an efficient alternative to deep models. DLinear decomposes time series into trend and seasonal components and applies direct multi-step linear projections, showing that simple methods can perform well when strong temporal regularities exist. LTBoost enhances linear models by combining multichannel linear regression with a residual modeling stage based on gradient-boosted decision trees, combining linear extrapolation with non-linear residual correction. These models demonstrate that well-designed linear systems remain competitive, particularly for long-horizon forecasting.

TimeKAN [96] introduces a frequency-aware framework based on Kolmogorov-Arnold Networks (KANs) [97] to model mixed-frequency time series. The model progressively decomposes input sequences using moving averages and applies cascaded multi-order KAN blocks as adaptive frequency filters. These filtered components are then integrated through a frequency mixing module to reconstruct the full prediction. This design achieves promising results across multiple long-term forecasting tasks, while using low computational resources.

There are also some researches exemplify modular hybrid architectures that combine multiple specialized components to address different aspects of the forecasting pipeline. MSGNet [98] performs Fourier-based scale transformation and uses graph

convolution to model inter-series dependencies, followed by temporal attention for intra-series interactions. LEDDAM [99] combines a trainable decomposition module with dual attention mechanisms—channel-wise self-attention for variable dependencies and a hybrid autoregressive-attention unit for temporal modeling. SimpleTM [100] combines classical signal processing with lightweight modeling by using a stationary wavelet transform for multiscale tokenization and enhancing linear self-attention with geometric algebra to improve efficiency and representational power. In each case, the architecture is constructed from domain-aware modules targeting specific forecasting subproblems such as decomposition, dependency modeling, and frequency adaptation.

Collectively, these models illustrate the potential of non-standard architectural strategies—grounded in theory, domain knowledge, or modular composition—to create competitive, interpretable, and often highly specialized forecasting solutions. As time series modeling continues to evolve, these alternative approaches provide valuable complements to mainstream deep learning frameworks.

■ 5 Design trends and research directions

While Section 4 reviews models based on their architectural backbones, this section looks across individual designs to highlight key trends that reflect shared priorities and innovations in deep time series forecasting. Building on the challenges outlined in Section 3, we examine how recent methods address these core issues.

Rather than focusing on specific models, we highlight common methodological directions that have gained traction across the literature. As summarized in Fig. 2, these directions frequently span multiple architecture types and serve as reusable design principles in modern forecasting pipelines. By understanding the motivations and techniques behind these shared patterns, we aim to provide a conceptual map that can guide future development in the field.

5.1 Multi-scale modeling

Multi-scale modeling has become a widely adopted approach in time series forecasting due to its natural alignment with the hierarchical structure of temporal patterns. Instead of relying on a single temporal resolution, multi-scale methods extract and process information at various granularities, from short-term fluctuations to long-term trends, allowing models to capture diverse temporal dynamics more effectively. This approach enhances the model's ability to recognize periodicity, transitions, and structural changes, making it particularly valuable for complex real-world sequences that span multiple time horizons.

This technique is particularly effective in addressing the challenge of temporal dependency. By operating across scales, models can disentangle and explicitly represent both local and global behaviors. Some methods achieve this through parallel convolutional branches with different receptive fields, capturing short-term details and long-term structures independently. Others construct multi-resolution inputs via downsampling and then use specialized modules to mix or align signals across these temporal scales. These strategies allow the model to flexibly represent periodicity, seasonality, and macro-level trends that would otherwise be difficult to capture within a uniform

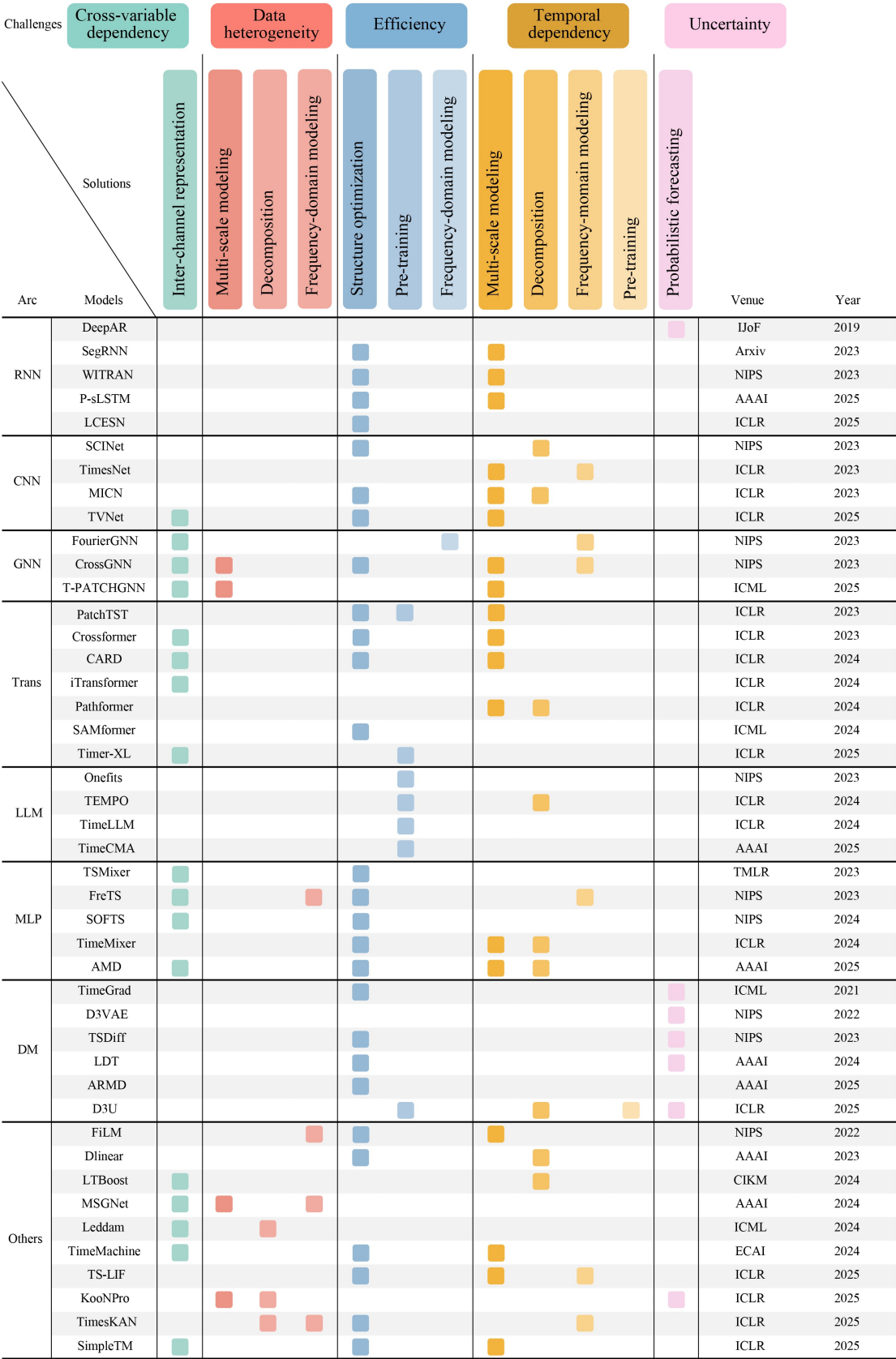


Fig. 2 Model–challenge mapping across design trends in time series forecasting. This diagram summarizes what key design principles representative deep learning models adopt to address major challenges in time series forecasting. It highlights patterns of methodological convergence, showing which design strategies are effective for tackling particular challenges

representation.

In addition, multi-scale modeling contributes to handling data heterogeneity, especially in the presence of noise or irregularity. Coarser resolutions tend to suppress noise and highlight more stable patterns, offering a robust foundation for learning cross-time relationships. Furthermore, in irregular multivariate time series, multiscale techniques can be adapted to perform patch-level alignment across variables, effectively mitigating issues such as asynchronous sampling and sequence length explosion. By transforming input sequences into temporally consistent patches, models can bypass pre-alignment procedures that are computationally expensive or fragile.

Beyond solving individual challenges, multi-scale representations also provide a modular interface for integrating other forecasting mechanisms. For example, attention mechanisms, graph structures, or decomposition methods can be combined with multiscale representations to enhance learning robustness and expressiveness. As a result, multi-scale modeling serves as both a technique and a design principle that underlies a diverse range of successful forecasting frameworks.

5.2 Decomposition

Decomposition is a widely used strategy in time series forecasting that aims to separate raw sequences into interpretable and structurally distinct components, most commonly trend, seasonality, and residual noise. This approach draws from classical time series analysis, where decomposing signals into additive or multiplicative elements has long served as a foundation for more stable and interpretable modeling. In modern deep learning frameworks, decomposition is reintroduced as a means of simplifying learning dynamics and improving performance.

A key benefit of decomposition is its ability to improve the modeling of temporal dependency. By isolating slow-changing trends or repeating seasonal patterns from noisy residuals, decomposition allows different model components to specialize in different temporal structures. Some methods apply dedicated encoders or linear projections to each component, while others integrate decomposition into the network backbone through soft routing, selective attention, or scale-specific parameter tuning. These designs enable models to capture long-term trends and short-term variations more effectively by reducing interference between conflicting signals.

Decomposition also plays a critical role in addressing data heterogeneity. Time series with mixed frequencies, nonlinear variations, or abrupt disturbances are notoriously difficult to model as a whole. By explicitly separating low-frequency and high-frequency components, decomposition helps isolate trend, seasonal, and residual signals, enabling models to better handle nonstationary behaviors and adapt to noisy or irregular inputs. In multiscale settings, decomposed representations can further inform the selection of temporal scales or guide the weighting of different components, resulting in more stable, adaptive, and context-aware forecasting performance.

Moreover, decomposition provides a foundation for interpretable forecasting pipelines, where different components of the input

contribute transparently to the final prediction. As such, it serves not only as a modeling aid, but also as a diagnostic tool for understanding temporal dynamics within complex datasets.

5.3 Frequency-domain modeling

Frequency-domain modeling has emerged as a flexible and powerful strategy in time series forecasting, offering both theoretical and practical advantages. By projecting raw sequences into the frequency domain, typically via Fourier transforms, models gain access to an alternative representation of temporal data that emphasizes periodic structures, long-range correlations, and dominant time scales. This transformation enables forecasting models to approach challenges from a spectral perspective, often simplifying modeling and amplifying key signals.

One of the key advantages of frequency-domain modeling is its ability to address temporal dependency modeling. Many temporal patterns, such as periodic changes, seasonal cycles, and trend shifts, are easier to detect in the frequency domain, where short-term and long-term components can be separated. Models that operate on frequency coefficients are able to extract global signals, separate trends from noise, and reduce interference across time scales more effectively. Some approaches convert sequences into spectral components for primary processing, while others use spectral decomposition as a preprocessing step to inform multiscale attention, routing, or mixing operations.

Frequency-based methods are also effective for handling data heterogeneity. Time series often contain mixtures of frequencies, abrupt disturbances, and inconsistent signal-to-noise ratios. By applying techniques in frequency domain, models can isolate stable frequency bands while suppressing noisy or high-variance components. This enables more reliable learning, especially in long-horizon forecasting, where noise accumulation can degrade performance. Moreover, frequency decomposition is particularly useful in irregular or multi-variable time series, where aligned periodicities may emerge at different levels.

Combined with innovative design ideas, some models even use this method to improve efficiency, like FourierGNN applies this approach to reduce cost by replacing time-domain graph convolutions with matrix multiplications in the Fourier space, showing how frequency-domain operators can dramatically reduce complexity, while preserving expressive power. Such reformulations open the door to architectures that leverage frequency domain for more efficient learning and inference.

Noticeably, frequency-domain modeling is not bound to a single architecture, it can serve as a design tool, a preprocessing step, or an integral part of the model's backbone. Its adaptability allows it to be embedded into a wide range of forecasting systems, often enhancing both performance and interpretability.

5.4 Inter-channel representation

Inter-channel representation focuses on capturing dependencies among variables in multivariate time series. In many real-world forecasting tasks, different variables, whether sensors, financial indicators, or system metrics, interact over time. Their co-evolution

introduces rich structures that cannot be effectively modeled by treating each channel independently. Learning effective inter-variable representations has thus become a central strategy for improving multivariate forecasting.

This approach directly maps with the challenge of cross-variable dependency modeling. Instead of modeling each channel in isolation, inter-channel representation enables the model to recognize when signals across variables are correlated, synchronized, or causally linked. This is particularly important in multivariate setups where the future of one series may depend heavily on the patterns of others. By extending the modeling context to encompass these interactions, models can produce more informed and accurate predictions.

Techniques for modeling inter-channel dependencies are diverse. Some models leverage the strengths of their backbone architectures, for example, GNNs, which are naturally suited for capturing inter-variable relationships through graph structures. Attention-based mechanisms are also widely adopted, often with custom modifications to better capture cross-variable interactions. In addition, some approaches combine multiscale alignment with frequency-domain projections to extract richer insights from inter-channel dependencies.

This methodological flexibility shows that there is no single blueprint for inter-channel modeling. What these techniques share is a common goal: to extend forecasting models beyond univariate structure, treating multivariate time series not as collections of independent sequences, but as interdependent signals with shared dynamics.

5.5 Structure optimization

Structure optimization refers to architectural or algorithmic design choices that aim to reduce the computational cost, memory usage, and inference time of forecasting models. In deep time series forecasting, where long input sequences and large variable dimensions are common, improving efficiency is crucial not only for deployment in real-time systems but also for scalability during training.

This method directly tackles the challenge of efficiency by polishing model execution without compromising accuracy. Some approaches achieve this through the selection of inherently lightweight backbones, such as MLPs or linear models, which offer low computational complexity by design. Others begin with more expressive but costly architectures, like transformers or GNNs, and propose targeted optimizations to reduce their time and space complexity. These optimizations can be very diverse, ranging from token sparsification, dynamic routing, and low-rank approximations to acceleration modules that restructure information flow or reduce the number of operations required.

Designs often exploit the strengths and constraints of their underlying architectures. Transformer-based models, for instance, commonly address the quadratic cost of self-attention by incorporating projection modules, exponential smoothing layers, or memory-efficient conditioning strategies that preserve global context while improving scalability. Graph-based approaches reduce complexity by filtering edge connections, learning sparse or dynamic

graphs, or reparameterizing node interactions to retain expressiveness with linear or near-linear runtime. In sequence modeling, recurrent-style modules may be refactored into partially parallel forms, enabling faster forward passes while preserving the temporal inductive bias that makes recurrent architectures effective.

Across all these variations, structure optimization emerges as a pervasive consideration in modern forecasting research. It represents a balance between strong generalizability and resource efficiency, enabling models to scale to longer sequences, more variables, or larger deployment environments without sacrificing performance. As such, it forms a critical design dimension in nearly every new architecture proposed for time series forecasting.

5.6 Pre-training

Pre-training has become an increasingly influential method in time series forecasting, offering a powerful way to improve efficiency, particularly from the standpoint of data efficiency. By learning general temporal representations or sequence structures from large-scale datasets, models can transfer this knowledge to downstream forecasting tasks with limited supervision. This not only reduces the required labeled data but also accelerates convergence and minimizes training cost, especially in low-resource or domain-adaptation scenarios.

Most commonly, pre-training is employed in LLM-based architectures, where large language models are adapted to forecast time series through tokenization, embedding, and autoregressive generation mechanisms. These models often benefit from prompt-based fine-tuning, where learned prompt tokens guide the forecasting behavior while keeping the backbone frozen, dramatically reducing the number of trainable parameters. In addition, soft prompts or pre-learned embeddings can inject decomposed components (e.g., trend, seasonality) to enhance sample efficiency and reduce model variance during adaptation.

While pre-training is most often associated with efficiency gains, it can also contribute to improved temporal dependency modeling. For instance, when combined with generative models, pre-trained point predictors can offer strong initialization for capturing low-uncertainty components of the sequence, thereby providing reliable conditions for modeling more complex or uncertain dynamics via downstream modules such as diffusion-based predictors. This dual-stage structure enables better separation of deterministic and stochastic elements, leading to more accurate and stable long-range forecasts.

Beyond LLMs, pre-training is also being explored in traditional architectures like transformers, where self-supervised learning on large corpora of time series data enables zero-shot or few-shot forecasting with competitive performance. This growing trend points toward a future where pretrained forecasting models could serve as general-purpose backbones across diverse time series domains.

5.7 Probabilistic forecasting

While most modern time series models focus on point forecasting, there is growing interest in methods that can capture predictive uncertainty. Probabilistic forecasting seeks to model the full distribution of future values rather than producing a single estimate,

which is essential for decision-making in high-stakes or stochastic environments. This approach is especially useful in scenarios where rare events, outliers, or randomness can greatly affect outcomes.

Addressing the challenge of uncertainty estimation, probabilistic forecasting naturally aligns with generative modeling approaches. Unlike deterministic predictors that produce fixed outputs, generative models can sample from learned distributions, enabling a richer understanding of possible futures. Variational inference, diffusion processes, and autoregressive likelihoods offer different mechanisms to model uncertainty explicitly and adaptively. These models not only produce confidence intervals or probabilistic trajectories but also disentangle high-certainty and high-variance components in the temporal signal.

Generative diffusion models, in particular, have recently gained traction due to their ability to capture complex distributions. By simulating both forward and reverse transformations of time series data, these models can represent noisy and ambiguous dynamics more faithfully than traditional point predictors. Similarly, variational autoencoders (VAEs) and neural processes have been adapted to learn latent representations that reflect the variability in temporal patterns, offering scalable and flexible uncertainty modeling.

In practice, probabilistic forecasting has demonstrated significant advantages in applications ranging from demand prediction to anomaly detection. As the field continues to mature, future research is likely to push further into generative field, developing more expressive priors, diffusion-based samplers, and hybrid structures that balance predictive accuracy with uncertainty awareness.

6 Future opportunities

As deep learning for time series forecasting continues to mature, recent trends suggest a shift from purely architectural innovations toward more principled and challenge-driven design. The methods surveyed in this paper, such as multi-scale modeling, frequency-domain operations, decomposition, and structure-aware optimization, point to an emerging toolkit of techniques that transcend specific

model types. These strategies will likely remain foundational in future developments, but several broader opportunities warrant continued exploration.

A particularly notable trend is the growing interest in leveraging models originally developed outside the time series domain, especially large pretrained models and modular frameworks designed for language or vision tasks [67]. This shift reflects a broader movement toward cross-domain generalization, where methods like prompt tuning, latent diffusion, and multi-modal embeddings are repurposed for forecasting tasks. While these models offer strong representation power and sample efficiency, their direct application to temporal data often raises alignment issues. Challenges such as preserving temporal continuity, handling irregular intervals, or scaling to multivariate sequences must be addressed to fully realize the potential of these models in time series contexts. Future work should therefore focus on adapting general-purpose architectures with domain-specific enhancements that respect the unique properties of time-indexed data.

Beyond model design, an important future direction is to focus more on the time series data itself, exploring how processing, transformation, and representation can improve alignment with model assumptions and boost model performance [19,101]. Rather than treating the input as fixed, researchers are increasingly viewing time series as flexible signals that can be adjusted to highlight important structures or reduce complexity. As illustrated in Fig. 3, many recent models combine architectural modifications with series-specific processing methods, such as dimension transformation, rearrangement, feature engineering, or decomposition, demonstrating that smart data preparation can be just as impactful as model innovation. In this classification, models are grouped based on their primary approach to processing the input series, and we also identify whether they modify architectural components, such as introducing new techniques or redesigning key blocks. Future work should continue exploring how to design processing pipelines tailored to specific temporal patterns to enhance forecasting accuracy.

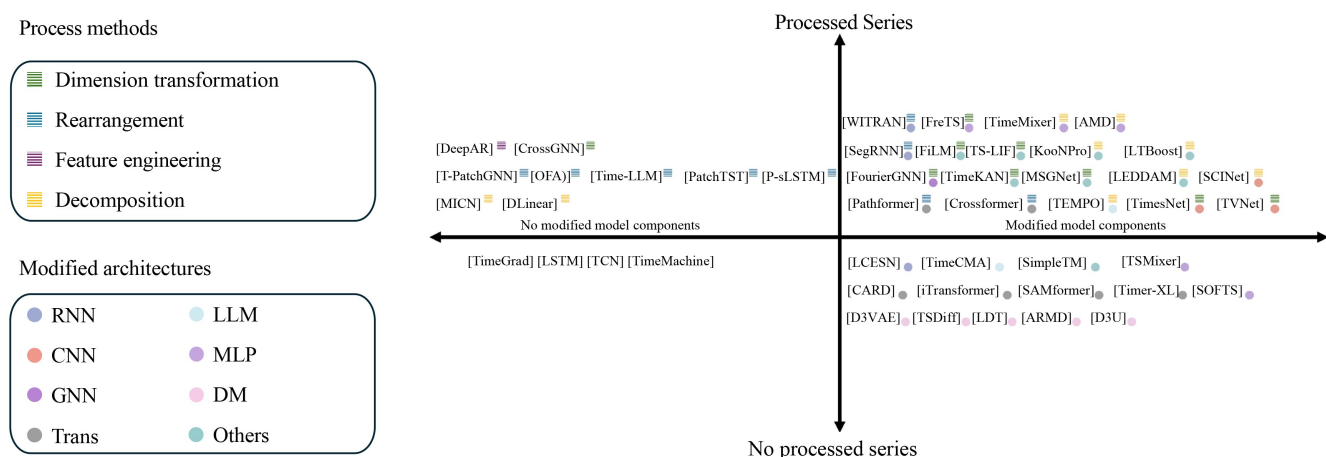


Fig. 3 Overview of design trends in time series forecasting. This figure categorizes recent deep learning models based on two binary dimensions: whether they process the input time series (vertical axis) and whether they modify model components (horizontal axis). Each model is annotated with its processing methods and the type of architecture it modifies. This 2×2 grid highlights four distinct design trends and reveals how methodological choices align with architectural innovation across the forecasting literature this paper explore.

Finally, future research should continue exploring how to build robust models under data scarcity, noise, and irregular sampling, conditions common in real-world deployments. Some recent research has already explored these situations, but more work is needed to benchmark their effectiveness in time series across diverse tasks and datasets.

Overall, the future of time series forecasting will be shaped not only by the development of new model architectures, but also by how well these models integrate with the unique properties of temporal data. With the field evolving, a holistic approach that balances architectural innovation with thoughtful data handling and principled design will be critical to building the next generation of forecasting models.

7 Conclusion

This survey presented a challenge-oriented synthesis of recent deep learning methods for time series forecasting. Rather than focusing solely on architectural classifications, we emphasized common design principles, such as multi-scale modeling, decomposition, frequency-domain operations, and pre-training, and analyzed how they address key forecasting challenges including temporal dependency, data heterogeneity, efficiency, cross-variable interactions, and uncertainty estimation.

In addition to providing a descriptive summary of recent work, we map each method to the core challenges it aims to solve, offering actionable insights into why particular strategies succeed and how they can be adapted across domains. Through our structured challenge categorization, methodological synthesis, and design trend analysis, this paper provides intuitive reference points for practitioners. Based on the characteristics of specific data and the requirements of the task, this challenge-method mapping offers a practical framework to guide model selection and assist practitioners in identifying suitable modeling strategies. Overall, we aim to help shift the focus from model inheritance to principled problem-solving, providing a solid foundation for both research advancement and applied forecasting practice.

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Competing interests

The authors declare that they have no competing interests or financial conflicts to disclose.

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